

Tony Nguyen

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EDUCATION

Vassar College

Poughkeepsie, NY

Bachelor of Computer Science | GPA: 3.95

- *Relevant Coursework: Data Structures & Algorithms, Software Engineering, Computer Systems, Data Science*
- *Awards: ASA DataFest 2025 - Best Data Visualization, MLH HackNights: Best use of Gemini API, NYCWiC: Top 2 Best Undergraduate Poster*

EXPERIENCE

Undergraduate Open-Source Research Fellow

Jun. 2026 – Aug. 2026

Vassar College | Prof. Josh de Leeuw

Poughkeepsie, NY

- Architected an end-to-end test automation suite using Playwright and BrowserStack, integrated into GitHub Actions CI/CD, and led a major Jest version upgrade—resolving the resulting breaking changes alongside several other dependency updates—while closing security gaps flagged by OpenSSF Scorecard.
- Redesigned a multi-agent LangGraph architecture by replacing shared context with isolated, task-specific routing, reducing per-execution inference costs 79%.

Full-Stack Software Developer Intern

Jun. 2025 – Sep. 2025

FPT Information Systems (FIS)

Hanoi, Vietnam

- Cut duplicate records by **35%** across a **500K+** row dataset by owning the master data table dashboard (**React.js**, **Ant Design**) from ambiguous requirements through production, redesigning the merge workflow end-to-end.
- Reduced data refresh latency by 40% by defining API contracts and integrating FastAPI endpoints to achieve near-real-time sync between a proprietary API and React state.

Computer Science Teaching Assistant

Aug. 2025 – May 2026

Vassar College

Poughkeepsie, NY

- Improved submission rates by **15%** across **90+** students in Data Science, Functional Programming, and Algorithms by designing targeted **Python/NumPy/Pandas** examples; reviewed **70+** Java and OCaml projects applying **OOP** principles, cutting recurring errors **20%**.

Engineering Makerspace Lab Assistant

Aug. 2024 – Present

Vassar Innovation Lab

Poughkeepsie, NY

- Resolved hardware/software failures for **200+** users across 3D printers and laser cutters; automated maintenance via **Python** scripts generating G-code test patterns and bed-level calibration sequences, eliminating manual test steps.

LEADERSHIP & CAMPUS ACTIVITIES

Outreach Chair

May. 2025 – Present

VC++

Poughkeepsie, NY

- Partnered with **10+** tech companies and alumni on mentorship and recruiting events, growing career programming participation **40%** and raising post-event satisfaction to **70%**.

PROJECTS

OBRE | *Android, Java, OpenCV, YOLOv12, ONNX, Firebase* [Link](#)

Feb. 2025 – Present

- Architected a modular, **object-oriented** real-time vision system (CameraX + **YOLOv12 ONNX** + ML Kit OCR) separating ingestion, inference, and output layers; implemented async processing to maintain stable throughput under variable workloads.
- Spearheaded end-to-end delivery from camera stream to on-device TTS, improving system isolation and maintainability across the full pipeline.

Aevol | *FastAPI, Next.js, PostgreSQL, Redis, Python asyncio, Docker* [Link](#)

2025

- Engineered a **distributed AI evaluation platform** with a Redis-backed priority queue using **Lua atomic task claiming** to prevent double-execution across horizontally scaled worker pools, with heartbeat monitoring, dead worker detection, and automatic re-queueing for fault tolerance.
- Built a pluggable evaluator system supporting exact match, embedding similarity, **LLM-as-judge**, and agent trace evaluation, with TF-IDF + KMeans failure clustering, regression detection across experiment versions, and a live **SSE dashboard** streaming real-time task progress.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript/TypeScript, OCaml, SQL, Swift

Tools: PostgreSQL, REST API, Git, GitHub Actions, Machine Learning, CI/CD, Docker

Frameworks: React.js, Node.js, Express.js, Bootstrap, Ant Design, FastAPI, Netlify